Fiscal Year:	FY 2022	**Task Last Updated:**	FY 09/15/2022
PI Name:	Romagnolo, Donato Ph.D.		
Project Title:	Space Environment and Epigenetics of Endocrine Regulation of DNA Repair and Cell Cycle in Mammary Epithelial Cells		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721	**Congressional District:**	3
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	01/04/2021	**End Date:**	04/30/2022
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 4/30/2022 per NSSC information (Ed., 1/4/2022).		
Key Personnel Changes/Previous PI:	During the reporting period, the contribution of 2 technical staff were added to the project to perform radiation experiments.		
COI Name (Institution):	Selmin, Ornella Ph.D. (University of Arizona)		
Grant/Contract No.:	80NSSC21K0301		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	RESEARCH PRODUCTS Research products of this project include studies of genes that may be epigenetically susceptible to space ionizing radiation (IR) and microgravity and how these changes affect endocrine regulation of genes involved in DNA damage response (DDR) and cell cycle control. The term epigenetics refers to changes in gene expression in the absence of DNA sequence modifications and includes CpG methylation, histone modifications (e.g., histone acetylation, methylation), and expression of non-coding RNAs (e.g., microRNAs). TYPE OF INVESTIGATION This is a Ground-based New Space Biology (NSB) proposal that investigates one of the major recommendations of the 2011 Decadal Survey related to the influence of the space environment on animal cell and molecular biology. This work is responsive to the Space Cellular and Molecular Biology 2018 NASA Omnibus Research Announcement soliciting research answers as to how the space environment causes epigenetic changes. Key elements of the space environment include IR and gravity (i.e., microgravity, hypergravity) which have been suggested to alter a variety of biological processes including endocrine systems (Sub-topic AB1-B). The central hypothesis of this proposal is that IR and microgravity alter epigenetically endocrine regulation of DDR and cell cycle control in mammary epithelial cells. The rationale for this hypothesis stems from our own and published evidence that 1) the estrogen receptor-alpha (ERalpha) activates epigenetically the BRCA1 gene, which participates with other DDR factors (e.g., ataxia telangiectasia mutated, ATM; ataxia telangiectasia and Rad-3 related, ATR) in regulation of DNA repair during the S and G2 phases of the cell cycle; and 2) conversely, activation of the aryl hydrocarbon receptor (AhR) prevents ERalpha-dependent activation of BRCA1. Because IR and microgravity regulate AhR activity, we will test if IR, microgravity, and their combination alter epigenetically the DDR and cell cycle regulation. SPECIFIC AIMS Specific Aims of this NSB project are to investigate in human mammary epithelial cells under endocrine stimulation the epigenetic effects of IR (Aim 1) and microgravity alone or in combination with IR (Aim 2) on expression of genes involved in DDR and cell cycle regulation. OUTLINE OF THE PLAN TO ACCOMPLISH THE SPECIFIC AIMS Aim 1: Human mammary epithelial cells (HMEC, MCF10A) will be cultured in 2D tissue culture dishes in the presence of estrogen to simulate the physiological exposure under normal gravity of mammary epithelial cells to ovarian estrogens. Cells will be exposed for various lengths of time (i.e., short-, 1, 2, and 3 days; medium-, 6, 12, 18 days; and long-term, 24, 36, and 48 days) to 3 doses (0.5, 1.0, and 2.0 mSv/day) of IR. Aim 2. Mammary epithelial cells (HMEC, MCF10A) will be cultured in 2D, or to simulate microgravity, in rotating wall vessel (RWV) bioreactor 3D cultures in the presence of estrogen without or with IR (0.5, 1.0, and 2.0 mSv/day) for various periods of time (i.e., short-, medium, and long-term). Determinations will include changes in: (i) mRNA and protein expression of BRCA1, ATM, ATR, ERalpha, AhR, and AhR nuclear translocator (ARNT) respectively by RT-PCR and Western blotting; (ii) CpG methylation at BRCA1, ATM, and ATR genes by methylation-specific RT-PCR ; (iii) levels of acetylated histone 3 (AcH3) and trimethylated lysine 27 at histone-3 at the BRCA1, ATM, and ATR genes by chromatin immunoprecipitation (ChIP) assay ; (iv) recruitment of DNA methyltransferase-1 (DNMT1) and DNMT3a, lysine-specific demethylase-1 (LSD1), histone deacetylase-1 (HDAC1), and PcG enhancer of zeste homologue 2 (EZH2) at the BRCA1, ATM, and ATR genes by ChIP assay; (v) genome-wide profile of genes targeted by the ERalpha and AhR by ChIP-sequencing; (vi) DNA double strand breaks by immuno-detection of gamma-H2AX foci; and (vii) cell cycle distribution by flow cytometry.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In humans, it is estimated tens of thousands of DNA damage lesions occur in each cell of the body per day as a result of normal physiological processes. This work has the potential to inform about the breast cancer risks to humans exposed on Hearth to various levels and duration to radiation.
Task Progress:	September 2022 Update: The Principal Investigator notes that a manuscript is in preparation to summarize results of the investigation. When published, the paper will be added to the Bibliography for this project. (Ed., 9/20/22) The main objectives of this project were to examine the impact of radiation and microgravity alone, or in combination, on regulation of gene expression of genes involved in repair of DNA damage; endocrine regulation; and xenobiotic response in mammary epithelial cells as a model of human breast cancer. Main accomplishments made during the reporting period of this project include: • We have established cell culture experiments with mammary epithelial cells to investigate the effects of microgravity on various markers of gene expression associated with activation of the aryl hydrocarbon receptor (AhR); • We harvested RNA for RT-PCR expression following short-time exposure to microgravity and measured expression of various genes including BRCA1, AhR, cyclin D1, and markers of AhR activity such as P450s (CYP1A1); • We set up experiments for short, medium, and long-term treatment of mammary epithelial cells with radiation; • We have completed gene expression measurements related to the short-term exposures (i.e., gravity and radiation); • We have completed cell proliferation measurements for short-term exposures; • We have completed protein and gene methylation experiments for short-term exposures; • Collected samples related to medium term; • Completed long-term radiation experiments; • Analyzed changes in gene expression; • Performed DNA CpG methylation experiments.
Bibliography Type:	Description: (Last Updated:)